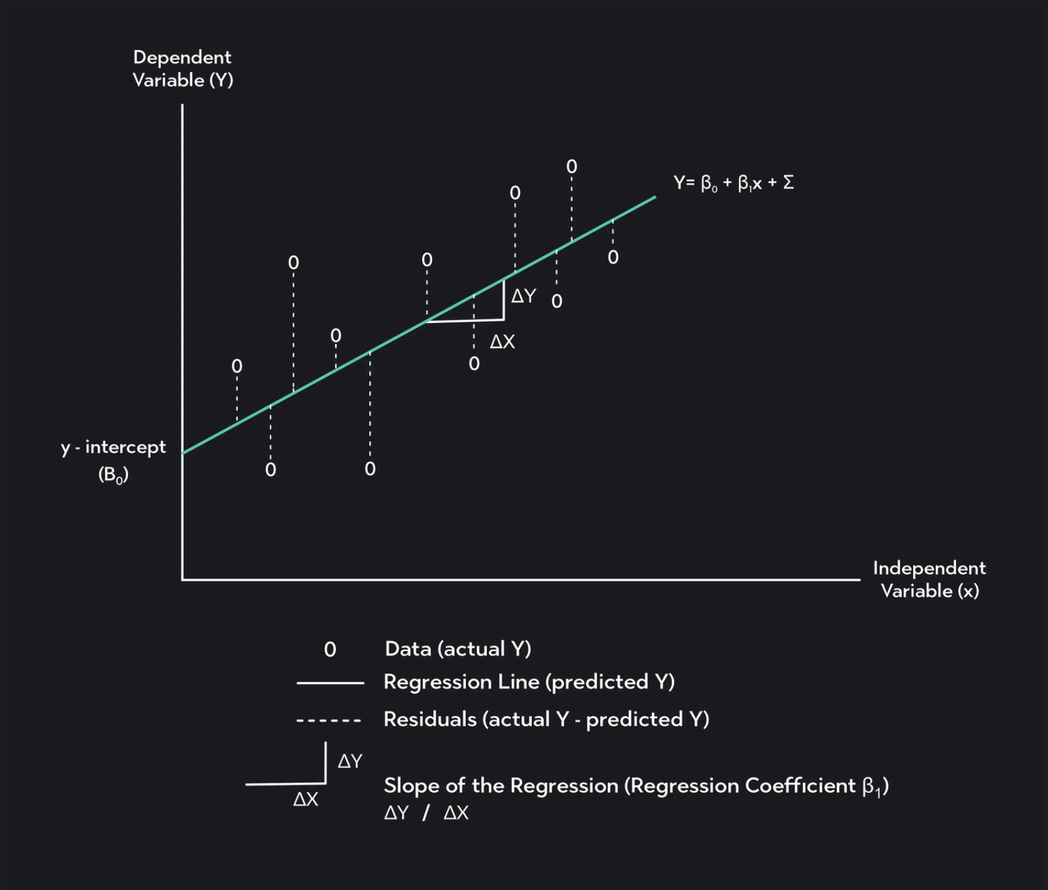
Linear Regression

Predicting store revenue with the Rossmann dataset

MLDS



INTRO

What we'll go through today

01

Overview of the data

1,115 drugstores, 2.5 years of daily sales

02

Linear regression

Fitting a line, then a line with many inputs

03

Evaluating a model

Residuals, MAE, RMSE, R^2 and fair testing

04

Putting it into practice

Going through the notebook

INTRO

From raw data to a model



Inspect

Plot it and look at some descriptive stats

Clean

Decide which rows to drop

Split

Train on the past, test on the future

Fit

Find the line that best fits the training data

Evaluate

How far off is it on unseen days?

It's a loop. What we learn from evaluating sends us back to inspect again, and what we see while inspecting decides how we fit.

01

The data



Rossmann stores

1,115

drugstores across Germany

~1M

rows, one per store per day

2.5 yrs

Jan 2013 – Jul 2015



OUR GOAL

Predict how much a store will sell on a given day.

What's in each row

Column	Meaning
`Store`	Store ID (1–1115)
`DayOfWeek`	1 = Monday … 7 = Sunday
`Date`	The day
`Sales`	**Revenue that day (€) ← our target**
`Customers`	Number of customers that day
`Open`	1 = open, 0 = closed
`Promo`	1 = running a promotion
`StateHoliday`	a = public, b = Easter, c = Christmas, 0 = none
`SchoolHoliday`	1 = public schools closed

One row = one store on one day.

Sales is what we predict. The rest are candidate features.

But should we use all of them?

Keep that in mind.

Look before you fit



What should we drop?

About 17% of rows are closed days. We don't need a model for those, so we keep only days the store was open.



Which numbers aren't numbers?

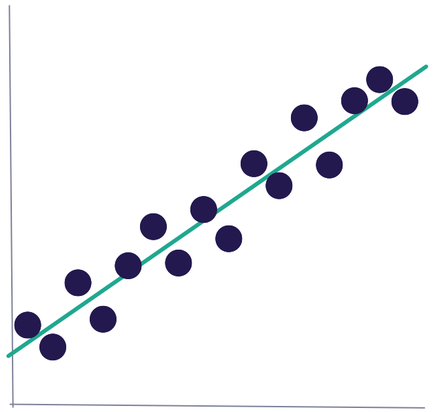
Store 1000 isn't “more” than Store 2, and Sunday (7) isn't “more” than Monday (1). These are categories.



Who's in the group?

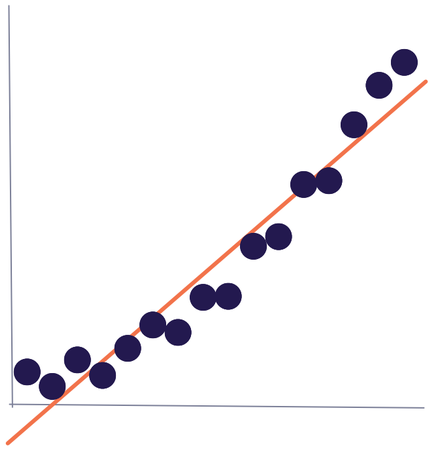
Only a few stores open on Sundays and holidays. A small, unusual group can make an average misleading.

Scatter first (feature against target)



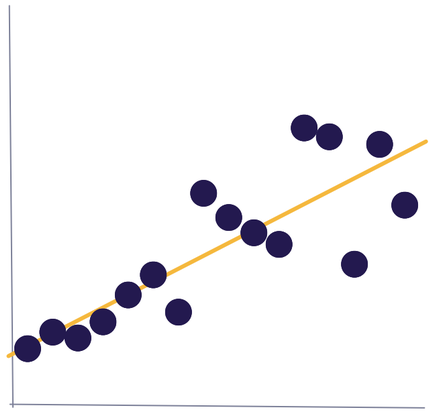
Straight band

A line fits. Go ahead.



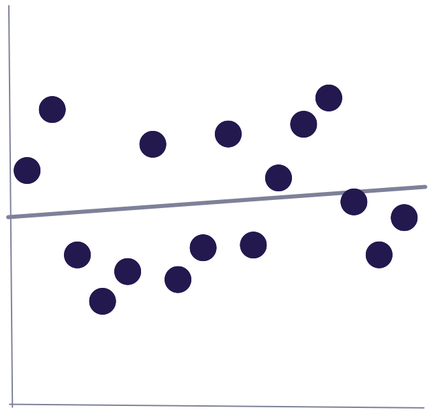
Curve

A line misses the bend



Fan shape

Trend is fine, but errors grow with x.



No pattern

This feature won't help much.

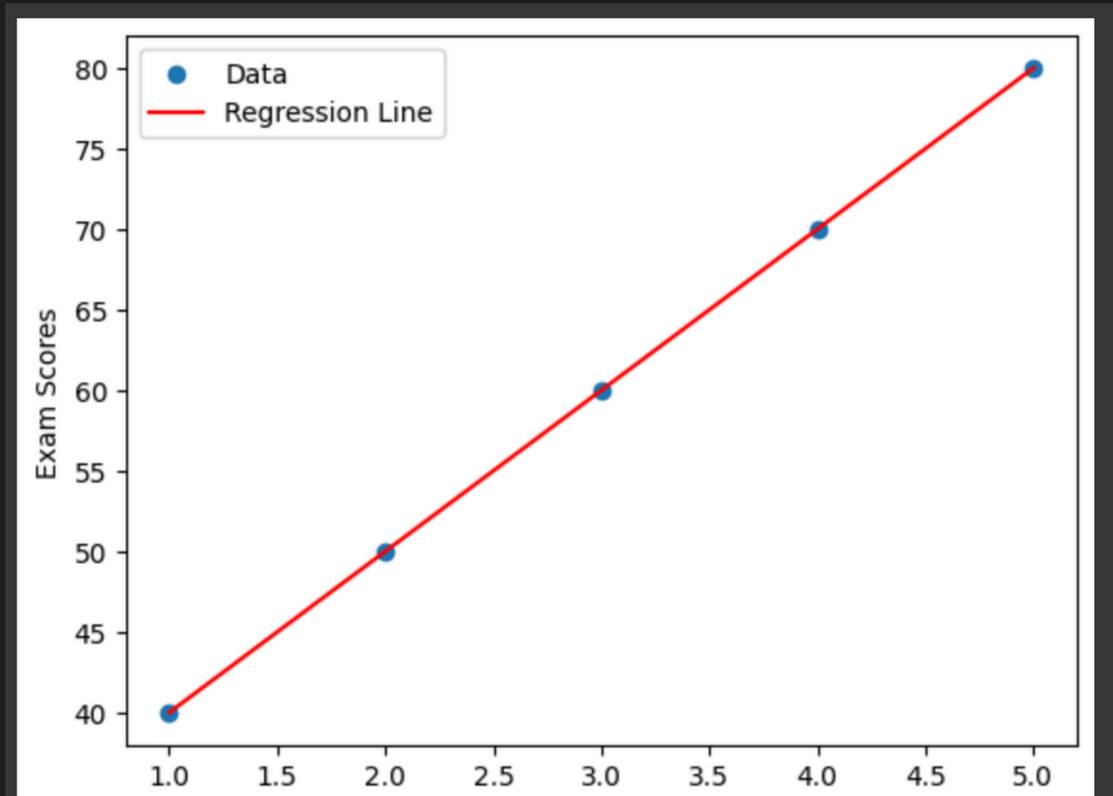
What inspecting tells the model

WHAT YOU SEE	WHAT IT MEANS FOR THE REGRESSION
A feature moves closely with sales	A good 1st feature for a simple regression
The scatter is a straight band	A linear model is a reasonable place to start
The spread fans out as x grows	Expect bigger errors on big stores, so check the residuals
The column is a category (day, store ID)	Encode it first (one-hot or target encoding)

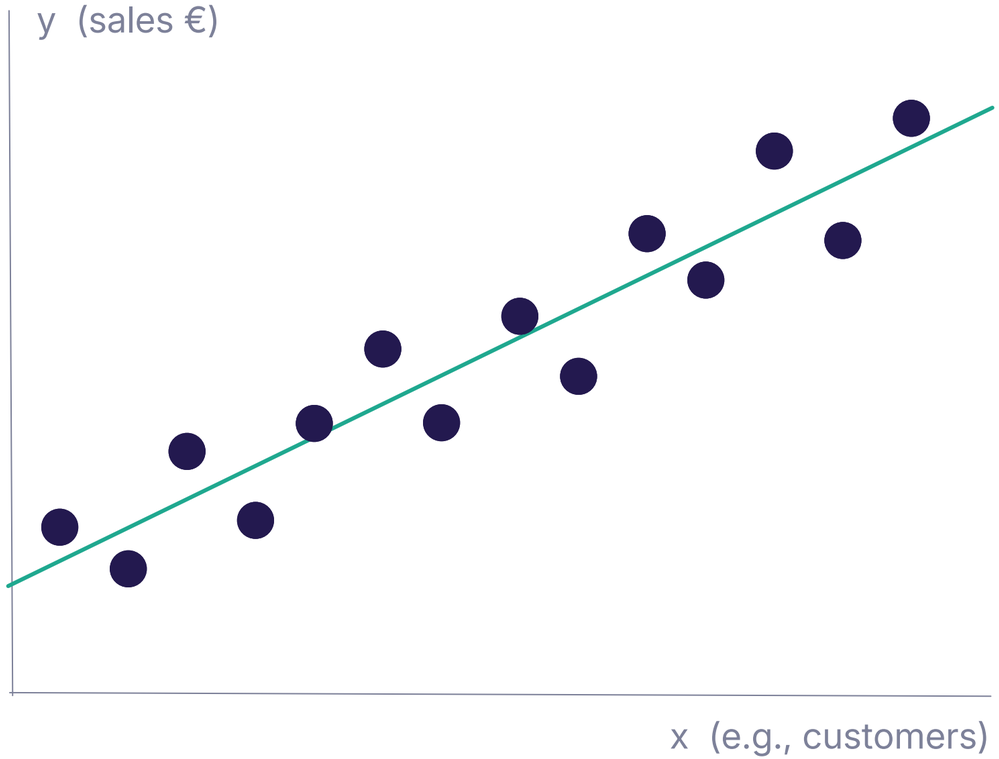
02

Linear regression

```
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt
# Sample data
X = np.array([1, 2, 3, 4, 5]).reshape(-1, 1)
Y = np.array([40, 50, 60, 70, 80])
# Create a linear regression model
model = LinearRegression()
# Fit the model to the data
model.fit(X, Y)
# Make predictions
Y_pred = model.predict(X)
# Plot the data and the regression line
plt.scatter(X, Y, label='Data')
plt.plot(X, Y_pred, color='red', label='Regression Line')
plt.xlabel('Hours of Study')
plt.ylabel('Exam Scores')
plt.legend()
plt.show()
```



The idea: draw the best fitting straight line



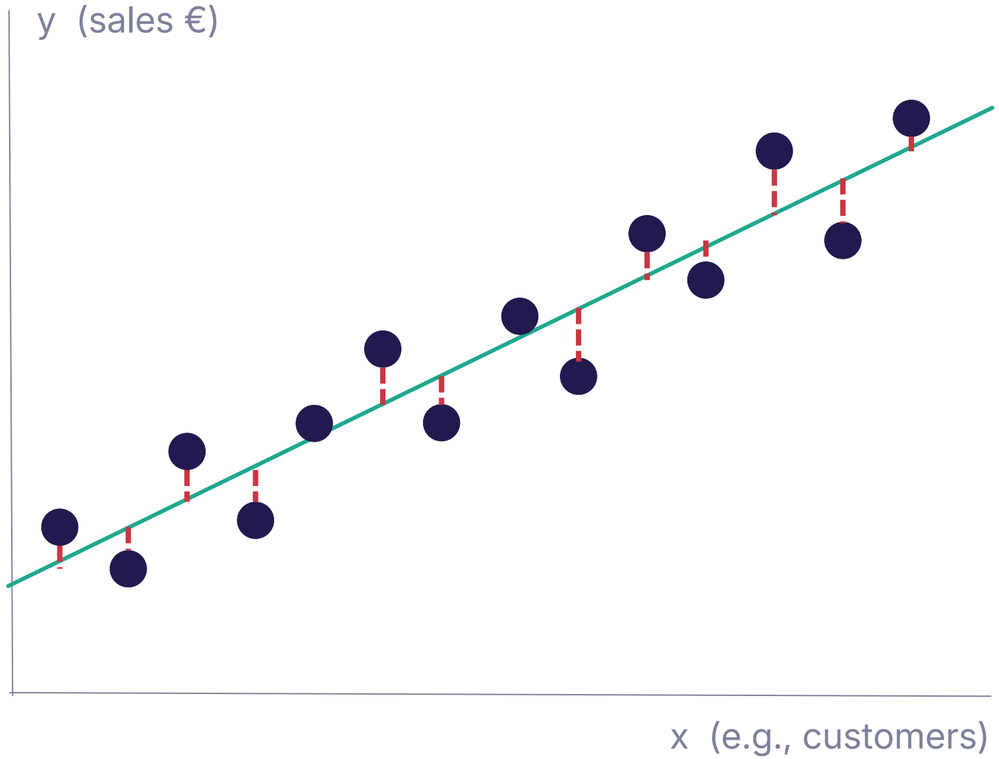
$$\hat{y} = b_0 + b_1 x$$

b_0 — **intercept**
The prediction when x = 0.

b_1 — **slope**
How much the prediction changes when x goes up by 1.

$\hat{y}$ — **prediction**
The line's guess for y.

How the line gets chosen

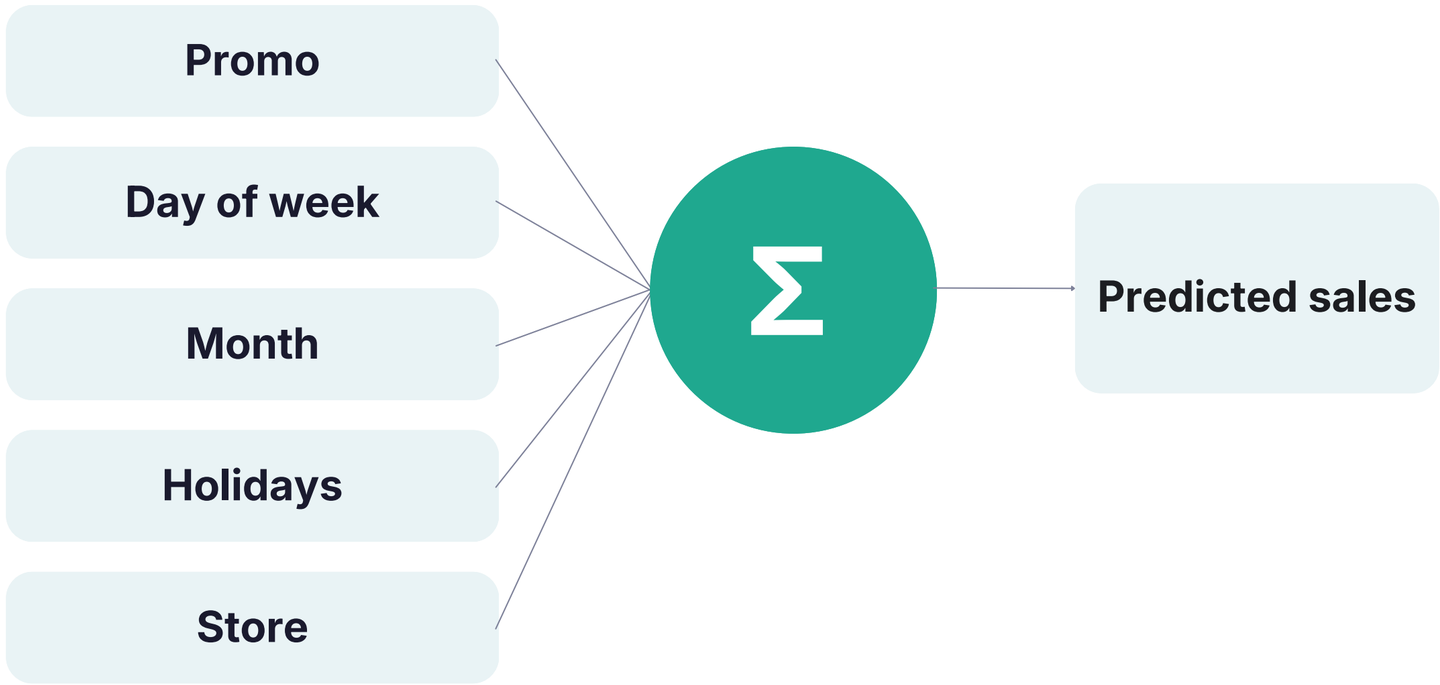


Residual = actual − predicted
The vertical gap between each point and the line (the dashed lines).

Least squares
Pick the b_0 and b_1 that make the sum of squared residuals as small as possible.

Many inputs: multiple linear regression

$$\hat{y} = b_0 + b_1x_1 + b_2x_2 + \ldots + b_kx_k$$



Reading a coefficient

b_i = how much the prediction changes when x_i goes up by 1, **holding everything else constant.**

Turning categories into numbers

One-hot encoding

Day	Tue	Wed	Thu	Fri	Sat	Sun
Mon	0	0	0	0	0	0
Tue	1	0	0	0	0	0
Sat	0	0	0	0	1	0
Sun	0	0	0	0	0	1

One 0/1 column per category. Monday gets dropped and becomes the baseline (all zeros), so each day's coefficient means “compared to Monday.”



Target encoding

1,115 store IDs would mean 1,114 columns. Instead, we can replace each ID with that store's average sales in the **training data** only.

What a linear model assumes

Linearity
The relationship is roughly a straight line.
Red flag: residuals curve

Independence
One row's error doesn't predict the next one's.
Red flag: errors come in streaks over time

Constant spread
Errors are about as big everywhere.
Red flag: a fan shape as predictions grow

Normal-ish residuals
Errors cluster around 0, symmetric.
Red flag: long tails or lopsided errors

03

Measuring a model

How far off are we, and compared to what?



Every metric starts with the error

€8,000
actual

−

€7,500
predicted

=

€500
error (residual)

Why not just average the errors?

Monday: +€1,000 too low. Tuesday: −€1,000 too high. Average error = €0.
Looks perfect, but both predictions were wrong. Positive and negative errors cancel, so we drop the sign or square it.

Four ways to summarize the errors

MAE

Mean Absolute Error

Average error, ignoring the sign

“On average we're off by €___”

Lower is better

MSE

Mean Squared Error

Square every error, then average

Units are €², hard to read

Lower is better

RMSE

Root Mean Squared Error

Square root of MSE

Back in €; big misses count extra

Lower is better

R^2

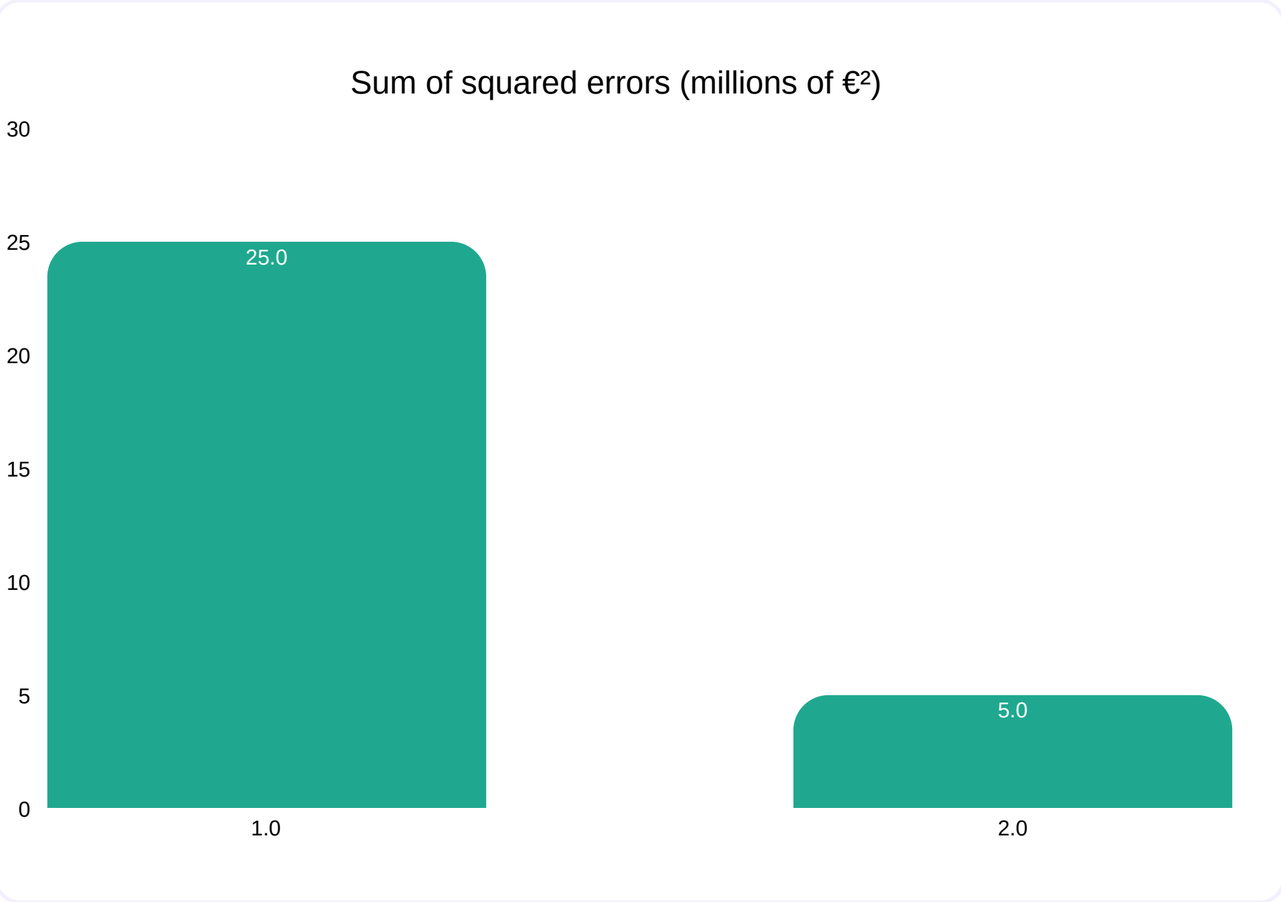
R-squared

Share of the variation in sales the model explains

1 = perfect, 0 = no better than guessing the average

Higher is better

Why RMSE punishes big misses



Both scenarios miss by €5,000 in total. Squaring makes the single big miss count 5× more.

Rules of thumb

- RMSE is always ≥ MAE
- RMSE much bigger than MAE → a few really big mistakes

Test on days the model has never seen



Why not shuffle randomly?

A random split could train on June and "test" on May

Split by date instead

In real life we forecast forward, so we train on the past and test on the most recent months.

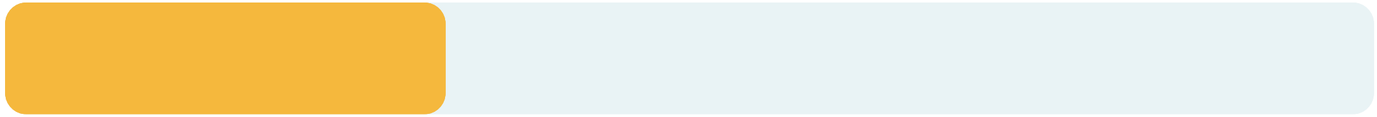
How good is good? Beat a baseline

A model is only as good as the simple guess it beats.

Always compare against something you could do without a model.

HOW MUCH OF SALES EACH APPROACH CAPTURES (ILLUSTRATIVE)

Guess the overall average every day · $R^2 = 0$ (by definition)



Guess each store's usual average · The baseline to beat



Your regression model · Only impressive if it goes further

